White Blood Cell Classification

IMA205 Challenge 2026

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Abstract

This report describes our approach to the IMA205 white blood cell (WBC) classification challenge. The task is to classify microscopy images into 13 cell types, evaluated with macro-averaged F1. The main difficulty is a severe class imbalance: the rarest class has only 11 training samples while the most common has over 13,000. We use a pretrained ConvNeXt-Base network and fine-tune it in two phases. Images are preprocessed with an adaptive median filter to remove impulse noise before being fed to the network. To handle the imbalance we apply a weighted cross-entropy loss with label smoothing. We also use geometric data augmentation during training and eight-view test-time augmentation at inference. Our model achieves a macro-averaged F1 of 0.7432 on the validation set and 0.7675 on 30% of the test set.

1 Introduction

White blood cells play a central role in the immune system and their morphology is routinely analysed in clinical practice to help diagnose infections, leukaemia, and other blood disorders. Automating this analysis could help pathologists work faster and more consistently, which is the motivation behind this challenge.

Each image must be assigned to one of 13 classes corresponding to different cell types and maturation stages. The evaluation metric is the macro-averaged F1 score, which computes the F1 independently for each class and then averages them with equal weight. As a result, correctly identifying a rare class contributes just as much to the score as correctly identifying a common one, which makes the task considerably harder than optimising for plain accuracy.

The two main challenges are (1) a very large class imbalance, with a ratio of roughly 1183:1 between the most and least frequent classes, and (2) high visual similarity between morphologically related cell types, particularly between different maturation stages of the

same cell families.

2 Dataset

The training set contains 28,901 images and the test set contains 9,634 images. All images are 368×368 pixels in RGB. The cells are well-centred by the acquisition system.

Table 1 shows the class distribution in the training set. The imbalance is extreme: *SNE* accounts for 13,015 samples while *PLY* has only 11. Several classes (PMY, PC, PLY) have fewer than 120 samples, which makes them very difficult to learn reliably.

Table 1: Training set class distribution.

Class	**Count**	**Fraction (%)**
SNE	13,015	45.0
LY	8,101	28.0
MO	2,746	9.5
BL	2,012	7.0
EO	861	3.0
MY	441	1.5
BA	415	1.4
BNE	391	1.4
VLY	366	1.3
MMY	360	1.2
PMY	114	0.4
PC	68	0.2
PLY	11	0.04
Total	**28,901**	**100**

We split the training data into a training subset (80%) and a validation subset (20%) using a stratified shuffle split with a fixed random seed. Stratification ensures that every class, including PLY with only 11 samples, is represented proportionally in both subsets.

3 Preprocessing

3.1 Adaptive median denoising

Many images in the dataset contain impulse noise (salt-and-pepper artefacts), which appears as isolated pixels with values that are very different from their neighbours. This

type of noise is common in microscopy due to sensor imperfections and digitisation. Left uncorrected, these artefacts can interfere with the fine-grained texture features that the network needs to distinguish cell types.

To remove this noise, we apply an adaptive median filter before any other processing. For each image we first compute an impulse noise score, defined as the mean absolute difference between each pixel and its 3×3 median neighbourhood. A high score indicates a lot of high-frequency impulse noise. Based on this score, we select the median filter kernel size as follows:

Table 2: Adaptive median filter kernel selection.

Noise score	**Filter kernel**
< 10	No filtering
$10 \leq s < 20$	3×3
$20 \leq s < 40$	5×5
≥ 40	7×7

Clean images receive no filtering, which avoids blurring meaningful texture detail unnecessarily. Only genuinely noisy images are processed, and the kernel size is proportional to the severity of the noise. A median filter is chosen over a Gaussian blur because it preserves edges — important for nuclear membrane sharpness — while effectively removing isolated noisy pixels. Figure 1 shows representative examples at each noise level.

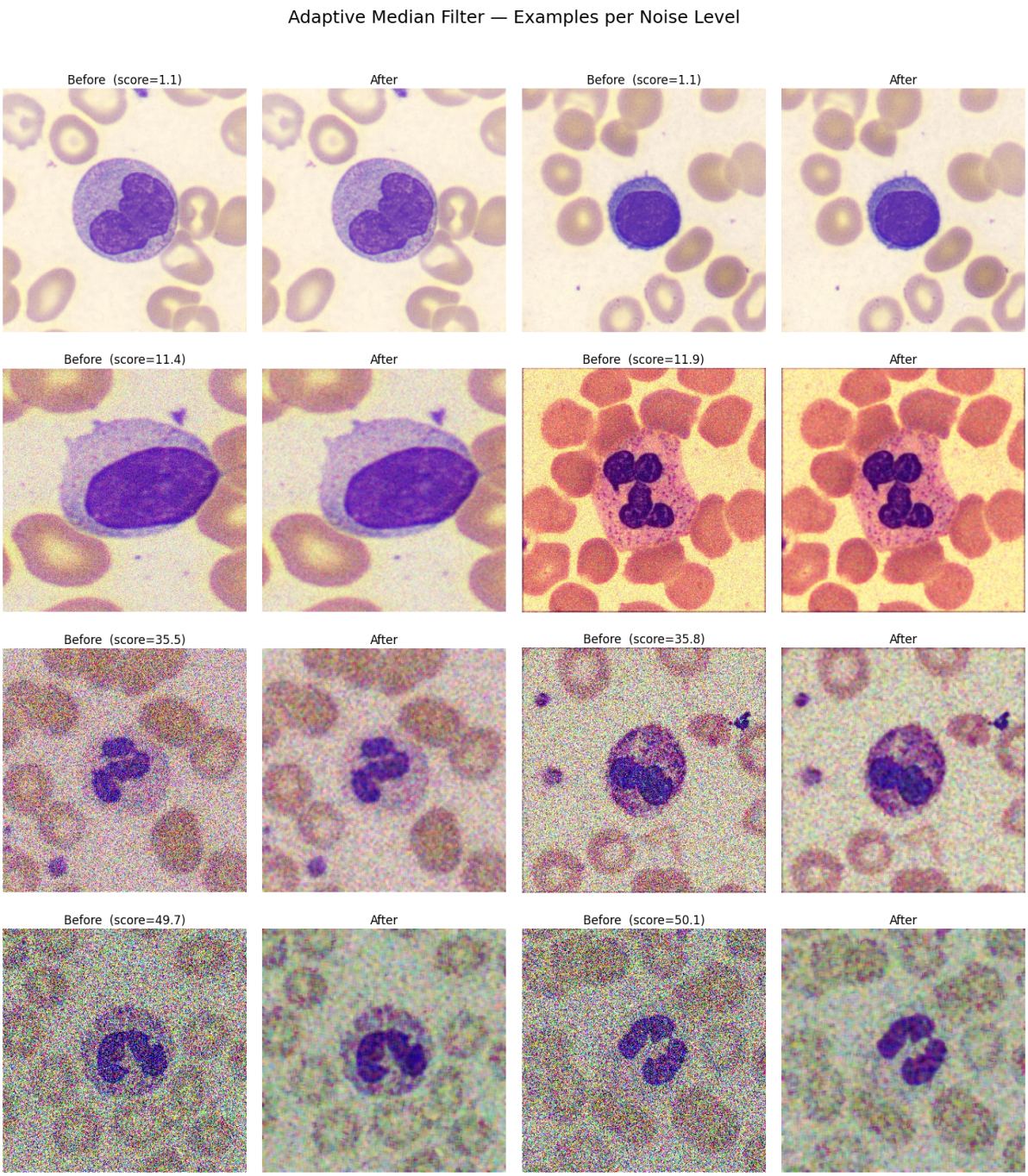


Figure 1: Examples of the adaptive median filter at each noise level. Each row corresponds to one noise bracket. Within each row, two before/after pairs are shown. Clean images (top row) are left unchanged. Noisier images receive progressively larger kernels, removing impulse artefacts while preserving cellular structure.

3.2 CenterCrop

After denoising, each image is centre-cropped to 300×300 pixels. The original 368×368 frame has a roughly 34-pixel background border on each side that carries no discriminative information, since the cells are already centred by design. Cropping removes this border

without any resizing, so the full resolution of the cell region is preserved and no interpolation artefacts are introduced.

3.3 Normalisation

Pixel values are normalised using the mean and standard deviation of the ImageNet dataset ($\mu = (0.485, 0.456, 0.406)$, $\sigma = (0.229, 0.224, 0.225)$). This is required because the backbone was pretrained on ImageNet with these statistics.

4 Model Architecture

4.1 ConvNeXt-Base

We use ConvNeXt-Base [1] as our backbone. ConvNeXt is a purely convolutional architecture that revisits the design of standard ResNets by incorporating ideas from vision transformers: depthwise convolutions with large kernels, inverted bottleneck blocks, the GELU activation function, and layer normalisation. Despite having no attention mechanism, it matches or exceeds transformer-based models on many benchmarks while retaining the translation equivariance and local processing properties that make CNNs well-suited for cell image analysis.

The Base variant has 87 million parameters and produces 1024-dimensional feature vectors. The classification head is a global average pooling layer followed by layer normalisation and a linear layer that maps the features to 13 class logits.

4.2 Pretrained weights

We use the weights `convnext_base.fb_in22k_ft_in1k` from the `timm` library. These weights were obtained by first training on ImageNet-21k (14 million images, 21,841 categories) and then fine-tuning on ImageNet-1k. The ImageNet-21k pretraining exposes the model to a much larger variety of textures, colours, and shapes than 1k-class training alone, producing feature detectors that transfer more effectively to new domains. All pretraining data is exclusively from ImageNet, which satisfies the competition constraint.

5 Training

5.1 Two-phase fine-tuning

We fine-tune the model in two phases.

Phase 1 (head only, 5 epochs). All backbone parameters are frozen and only the

linear classification head is trained, using a learning rate of 2×10^{-3}. The motivation is to avoid catastrophic forgetting. If we apply a random head's large gradients directly to the pretrained backbone from the first iteration, the features accumulated during pretraining are likely to be disrupted. Training the head first allows it to align its weights with the 13-class task so that, once the backbone is unfrozen, it receives informative rather than noisy gradient signals.

Phase 2 (full model, up to 30 epochs). All parameters are unfrozen and the full network is trained jointly with a lower learning rate of 1×10^{-4}. The lower rate ensures the backbone weights evolve gradually from their pretrained values. To reduce peak GPU memory usage during the backward pass, we enable gradient checkpointing, which trades a small amount of extra computation for a significant reduction in activation memory. Training stops early if the validation macro-F1 does not improve for a number of consecutive epochs (in the notebook, a patience of 5 was tested, but we found it is not enough, so we resumed training to obtain an even better model and interrupted the code later to avoid retraining from scratch; this is why one of the cells has a KeyboardInterrupt error).

5.2 Optimiser and schedule

We use AdamW [2] with a weight decay of 10^{-4}. AdamW decouples the weight decay from the adaptive gradient update, unlike standard Adam with L2 regularisation, which makes the regularisation more consistent across parameters with different gradient scales.

The learning rate follows a cosine annealing schedule in both phases. It decreases smoothly from its initial value to zero following a half-cosine curve, which lets the model take larger steps early in training and progressively finer steps as it converges.

5.3 Loss function

We use cross-entropy loss with two modifications to address the class imbalance.

Class weights. Each class c is assigned a weight

$$w_c = \frac{1}{\sqrt{n_c}}, \tag{1}$$

where n_c is the number of training samples in that class. The weights are normalised so that they sum to the number of classes. Using the square root rather than the raw inverse frequency is intentional: raw inverse weighting would give PLY a weight roughly 1183 times that of SNE, which would make training unstable. The square root compresses this ratio to about 34, which up-weights the rare classes meaningfully while keeping the

loss well-behaved. Weights are computed from the training split only to prevent any information leak from the validation set.

Label smoothing ($\varepsilon = 0.1$). Hard one-hot targets are replaced with $1 - \varepsilon$ for the correct class and $\varepsilon/(C-1)$ for all others. This prevents the model from becoming overconfident on the majority classes that dominate the early training batches and provides a mild regularisation effect.

6 Data Augmentation

During training, the following augmentations are applied on-the-fly after the denoising and centre crop. Validation and test images receive no augmentation beyond the crop and normalisation.

Table 3: Augmentations applied during training.

Transform	**Probability**	**Rationale**
Horizontal flip	0.5	WBC images have no meaningful left–right orientation.
Vertical flip	0.5	Same reasoning as horizontal flip.
Random 90° rotation	0.5	Cells can appear at any orientation; 90° rotations preserve the pixel grid exactly with no interpolation.
Gaussian blur (kernel 3–7 px)	0.3	Simulates the mild focus variation present in the dataset, making the model more robust to slightly blurry images at test time.

7 Test-Time Augmentation

At inference, each test image is passed through the network eight times using deterministic transforms: the original crop, horizontal flip, vertical flip, 90° rotation, and all pairwise and triple combinations of these three. The eight resulting softmax probability vectors are averaged element-wise, and the class with the highest mean probability is taken as the final prediction.

These eight transforms form the complete set of symmetries of a square. Averaging over the full symmetry group makes the prediction invariant to 90° rotations and reflections by construction, rather than relying solely on the model to have learned this invariance from

training data. For rare classes where the model has seen few examples, this reduction in prediction variance is particularly beneficial.

8 Results

8.1 Quantitative results

Table 4 summarises the macro-averaged F1 at each stage of the pipeline.

Table 4: Macro-averaged F1 at each training stage.

Stage	**Macro-F1**
Phase 1 — head only, single view	0.5937
Phase 2 — full model, single view	0.7425
Phase 2 + 8-view TTA (validation)	0.7432
Test set (30% of test data)	**0.7675**

Table 5 shows the per-class F1 of the final model on the validation set.

Table 5: Per-class F1 on the validation set (Phase 2 + 8-view TTA).

Class	**F1**
SNE	0.9818
EO	0.9704
LY	0.9604
BA	0.9518
MO	0.9086
BL	0.9221
MY	0.6887
VLY	0.5931
MMY	0.5781
PMY	0.5714
PC	0.5833
PLY	0.5000
BNE	0.4521
Macro avg	**0.7432**

Classes with the most training data (SNE, EO, LY, BA, MO, BL) all exceed F1 = 0.90. The lowest scores are concentrated in two groups: rare classes with limited training data

(PLY, PC, PMY), and classes that are morphologically similar to their neighbours (BNE, MMY, VLY), as discussed below.

8.2 Confusion matrix analysis

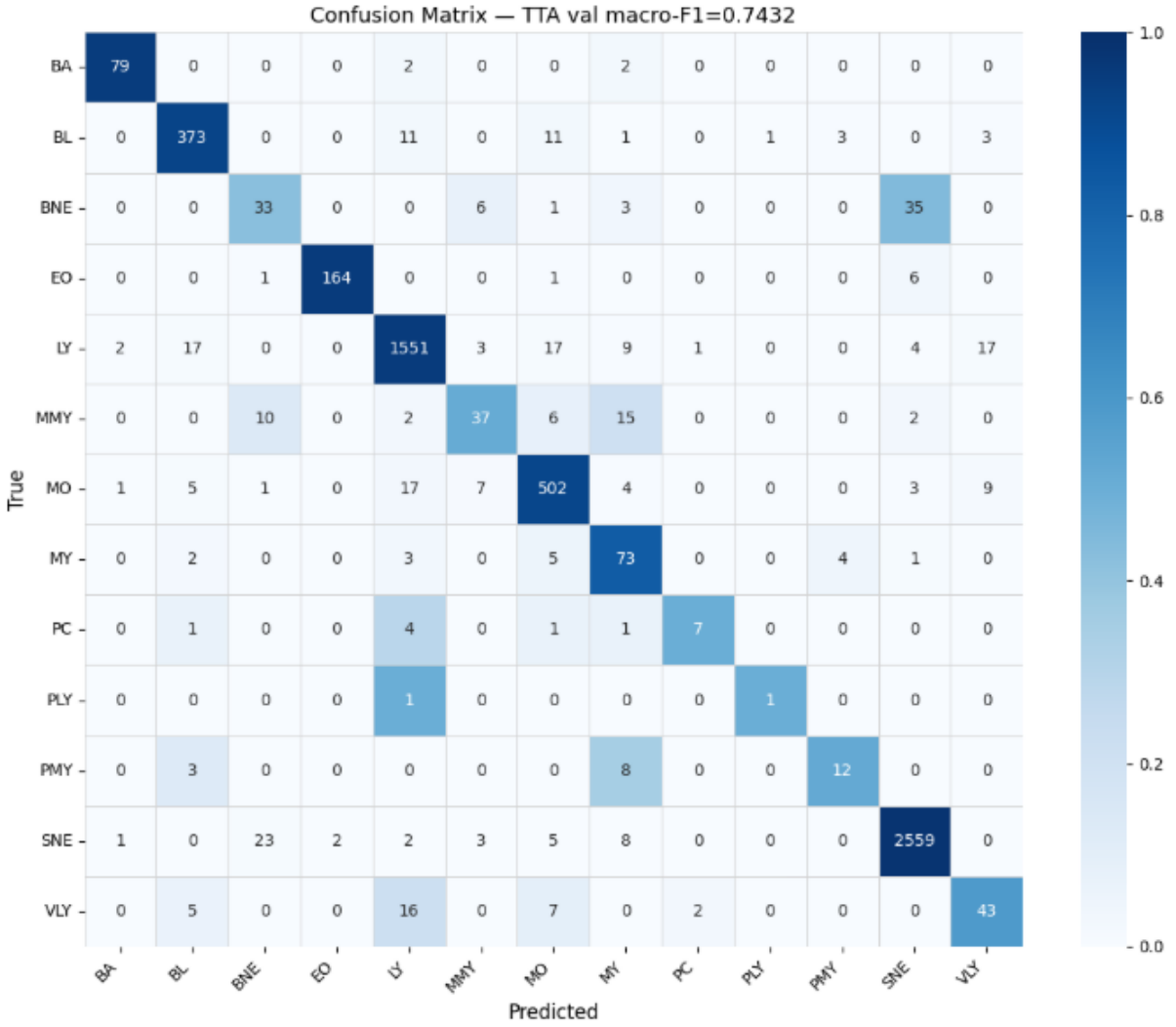


Figure 2: Normalised confusion matrix on the validation set (Phase 2 + 8-view TTA). Rows are true classes, columns are predicted classes. Colour intensity shows the fraction of true-class samples predicted as each class.

Several patterns are visible in Figure 2.

BNE ↔ SNE. Band neutrophils (BNE) are the most confused class, with a large share misclassified as segmented neutrophils (SNE). Both are neutrophils and share the same cytoplasm; the only difference is nuclear shape — BNE has a horseshoe or band-shaped nucleus while SNE has a clearly segmented, lobulated nucleus. The boundary between these two states is continuous in biology, so even human experts sometimes disagree on borderline cases. A potential improvement is to train a dedicated binary BNE/SNE

classifier and use it to post-process any prediction in this pair, since a classifier trained exclusively on these two classes can focus entirely on the shape of the nucleus without being distracted by the other 11 classes. Gathering more BNE examples should also help.

Maturation stage confusions: PMY, MY, MMY. Promyelocytes (PMY), myelocytes (MY), and metamyelocytes (MMY) are consecutive stages of the same maturation pathway. Their morphology changes gradually, making the boundaries between adjacent stages ambiguous. The confusion is mostly between neighbouring stages (PMY misclassified as MY, MY misclassified as MMY and vice versa) rather than between distant stages, which confirms that the model has learned the overall ordering.

VLY $\leftrightarrow$ LY. Variant lymphocytes (VLY) are occasionally confused with regular lymphocytes (LY). VLY cells are reactive lymphocytes with larger, irregularly shaped nuclei and more cytoplasm, but this distinction can be subtle. The fact that VLY has only 366 training samples likely contributes to the confusion. Gathering additional VLY examples, would be the most direct fix.

9 Conclusion

We presented a pipeline for 13-class WBC classification based on ConvNeXt-Base pretrained on ImageNet-21k. The method combines adaptive median denoising to remove impulse noise from raw images, two-phase fine-tuning to preserve pretrained features, a weighted cross-entropy loss with label smoothing to handle the severe class imbalance, geometric augmentation to improve generalisation, and eight-view test-time augmentation to reduce prediction variance. The model achieves a macro-averaged F1 of 0.7432 on the validation set and 0.7675 on the public test set. The main sources of remaining error are the morphological similarity between neutrophil subtypes (BNE vs SNE) and between adjacent maturation stages (PMY, MY, MMY), both of which are fundamentally constrained by limited training data and biological ambiguity at class boundaries.

References

[1] Z. Liu, H. Mao, C.-Y. Wu, C. Feichtenhofer, T. Darrell, and S. Xie. A ConvNet for the 2020s. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022.

[2] I. Loshchilov and F. Hutter. Decoupled Weight Decay Regularization. In *International Conference on Learning Representations (ICLR)*, 2019.